OPEN ITEMS CHECKLIST — part 6 (17 October 2026)

Changes since part 5 (16 October 2026). Stages 6a to 6c complete the Double-Number and Core Extension word sets and the `ENVIRONMENT?` query table. Every stage was assembled in all four builds (`SERIALPOLL` 0/1 by `UNITTESTS` 0/1) and with every test selector from 0 to 18.

Double-Number word set (stage 6a)

- ☑ **Added `D0=` , `D0<` , `D2*` , `D2/` , `M*/` , `2LITERAL` , `2VALUE`** , and `TO` now recognizes a `2VALUE` (it checks the behaviour cell at xt+3 for `DO2VALUE` ; a 2VALUE stores x2 at the lower address like `2CONSTANT`).
- ☑ **`M*/` method.** Take the magnitudes; form `|d1.lo|*|n1|` and `|d1.hi|*|n1|` with `UMUL32` and add them into a 3-cell product on an S frame; divide by `|n2|` in three 32/16 `UDIV32` steps (each quotient is under 65536); negate the result if an odd number of `d1` , `n1` , `n2` are negative. Symmetric division (truncates toward zero), `THROW -10` for a zero divisor. The method was checked against exact integers in a Python model (200000 random cases including the extremes) before it was written in assembler. Shadow entry `M*/W.1` .
- ☑ **Assembler unit tests** for `D0=` , `D0<` , `D2*` , `D2/` and `M*/` (including the divide-by-zero case) in the DArithmetic group (selector 4).
- ☑ **ANS test file `28_double_word.tests.fs`** built from the forth-standard.org Double-Number tests. Constants come from `D2/` , `M*/` and `MAX-D` instead of hard-coded values; typos fixed; the previously disabled blocks enabled. All tests pass in MAME. The `D.` / `D.R` test is visual (lines in pairs).

ENVIRONMENT? (stages 6a and 6b)

- ☑ **`CORE` , `CORE-EXT` , `DOUBLE` and `DOUBLE-EXT` answer.** A word-set query returns a value flag and then the recognized flag (`true true`). This matches the standard's table, where those queries return a flag, and gforth. An unknown name still returns a single `false` .
- ☑ **Section 24 tests cover the whole standard query table**, with `/HOLD` and `/PAD` checked against the standard minimums (34 and 84).

- ☑ The first version of the DOUBLE tests expected a single flag; the test was wrong, not the code.

Core Extension word set (stages 6b and 6c)

- ☑ **Added `.(`, `C"`, `SAVE-INPUT`, `RESTORE-INPUT` (6b), `S\"` and `CONVERT` (6c).** Found by auditing the dictionary against the standard word lists. Core had no gaps (`."`, `S"` and `ABORT"` only looked missing because of how their names are spelled in the source). `CORE-EXT` was briefly reported complete at 6b while `S\"` and `CONVERT` were still missing; the audit caught it.
- ☑ **`SAVE-INPUT`** saves the source address, source length and `>IN` (three cells and the count 3). **`RESTORE-INPUT`** restores `>IN` only if the source address and length match, and otherwise returns true, so it cannot restore across `REFILL` or into a different `EVALUATE` string.
- ☑ **`S\"`** translates `\a \b \e \f \l \m \n \q \r \t \v \z \" \\` and `\xhh` into PAD, then shares the `S"` code path (label `SQSTATE`). More than 255 characters throws -18. **`CONVERT`** is `>NUMBER` with no length limit.
- ☑ **ANS test file `29_coreext_strings.tests.fs`.** A `T{ ... >R ... R> ... }T` line at top level reset the target (`>R` leaves the value under the interpreter's return address); the test now uses a definition.
- □ **MAME confirmation of stage 6c** (`S\"`, `CONVERT` and the section 29 tests). Stage 6b was confirmed.

ROM layout and unit tests

- ☑ **`BASECODE` and `BASEDICT` lowered** to `$DC44` and `$D395` (were `$DCC4` and `$D458`). The interrupt-driven build no longer fitted below `INITCODE` once `S\"` was added. The dictionary ends exactly at `BASECODE` .
- ☑ Polled `BASECODE` is 8659 bytes (`$DC44` - `$FE16`), interrupt-driven 8978 (`$DC44` - `$FF55`), `BASEDICT` 2223 (`$D395` - `$DC43`). Padding below `INITCODE` : 397 bytes polled, **78 bytes interrupt-driven**.
- □ **Interrupt-driven headroom is 78 bytes.** The next addition to the base code needs another downward shift of `BASECODE` / `BASEDICT` (the spare ROM below `BASEDICT` is about 5 KB, but the unit test code sits there too) or a saving.
- ☑ **Unit test group 11 split.** In the interrupt-driven build the StrParse tests overran `BASEDICT` by 17 bytes. `REPLACES` , `SUBSTITUTE` , `CLR-REPLACES` , `SNAME` and `UNESCAPE` now run as group 18 (`TSTSTRPARSE2`); group 11 keeps the rest. Selector 0 calls both. `run all tests.sh` and `build all tests.sh`

need `18` added to their selector lists.

Documentation

- ☑ `ClaudeForth.docx` and preview PDF dated 17 October 2026: 13 new glossary entries and index entries (`D0=` , `D0<` , `D2*` , `D2/` , `M*/` , `2LITERAL` , `2VALUE` , `.(` , `C"` , `S\"` , `CONVERT` , `SAVE-INPUT` , `RESTORE-INPUT`), updated `TO` and `ENVIRONMENT?` entries, the glossary count (242 words; the earlier text said 227 while 229 entries existed), memory map table and ROM sizes, selector table (18), test-suite text, Section 5 items and the regenerated source appendix (stage 6c). Memory-map SVG and PDF regenerated.
- ☑ README (and README_ANS.md): sections 27 to 29, query-table and double notes, `>R` guidance; the Double-Number entries removed from the open items.
- □ **Timeline** on the title page not updated.
- □ Older checklists (parts 3 and 4) still list the Double-Number gap as open; this part supersedes them.